GCE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2024

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2024 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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GCE DIGITAL TECHNOLOGY

Chief Examiner's Report

Subject Overview

As with the 2023 series, direct comparison of the performance of this cohort with those of previous cohorts, either pre-COVID or post-COVID, would serve little purpose. Instead, this report highlights the strengths and weaknesses of the current cohort, avoiding any speculation about any lingering impact of Covid on the candidates.

However, it is important to repeat the main general points made repeatedly in previous Chief Examiner reports as these points still remain very pertinent as far as the current cohort is concerned.

- Candidates must be familiar with DT terminology and must use appropriate DT terminology in their responses.
- Where a question is framed around a context, candidates must address this context as this is an important element of the specification's AO2 assessment.
- Questions which are identified as assessing QWC are also a key element of the specification's AO3 assessment. The standard of response required for these questions is much higher, in respect of technical understanding, the accurate use of appropriate DT terminology and the Quality of Written Communication, as is indicated by the unit comments below and by the appropriate part of the mark schemes.

Assessment Unit AS 1 Approaches to Systems Development

Q1 **(a)** This question proved to be much more discriminating than intended. Only a small number of candidates could state three of the accepted reasons for the software crises. A significant number of candidates referred to budgeting which was given in the question stem.

(b) Most candidates were familiar with the user interface and could explain it.

(c) Most candidates could identify three resources. A significant weakness was a reluctance to use appropriate DT terminology typified by answers referring to 'money'.

(d) The overall level of response was high.

(e) Gantt charts were better understood than critical path analysis. A significant number of candidates used essentially the same words in describing both.

Q2 **(a)** Most candidates could identify three fact-finding methods in Part (i). The overall level of response to Part (ii) was very weak. The question required candidates to identify two appropriate documents but few candidates did.

(b) Many candidates were confused by the difference between functional and non-functional requirements and referred instead to essential and non-essential requirements.

(c) Part (i) discriminated very effectively. Many candidates could describe how a one level DFD was created from a context level DFD using the appropriate terminology and achieve three or four marks. An equal number of candidates were totally unfamiliar with the purpose of both. Many excellent Level 1 DFDs were produced for Part (ii). Common failures included the use of non-standard symbols, inappropriate process names and a lack of consistency with the given context level DFD.

Q3 **(a)** The overall level of response was weak with many candidates describing prototyping in general, often very inaccurately, but disregarding its use in the design of the user interface.

(b) The stem of the question referred to the waterfall model but the majority of candidates ignored this and described implementing a completed solution rather than the waterfall model's Implementation Stage as required. Few candidates gained any significant marks.

(c) This question discriminated as intended resulting in a wide range of marks. Candidates were stronger on the type of test data involved than its purpose. The example given deliberately suggested the wording likely to be found in a test plan, but few candidates availed of this resulting in many non-DT type answers.

(d) The overall level of response was very high with acceptance testing being better understood than system testing.

Q4 **(a)** Most candidates were appropriately familiar with a disaster recovery plan in Part (i) but many failed to use appropriate DT terminology. Part (ii) separated the candidature into two almost equal parts, those familiar with backups and those not.

(b) Many candidates correctly identified the type of maintenance and described it in the appropriate detail.

(c) The overall level of response was very disappointing. While many candidates could identify at least one example of technical documentation, most candidates could not relate technical documentation to its use in maintenance.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the agile and waterfall approaches in terms of development time, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(e) The overall level of response to this question was very high.

Q5 **(a)** While many candidates were familiar with the two terms, most struggled to explain them as required.

(b) As in Part (a), many candidates were familiar with the two terms, but most struggled to explain them in the context of the given algorithm as required.

(c) Only a small number of candidates were properly familiar with the two types of loop.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the use of inheritance in supporting code reusability, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Assessment Unit AS 2 Fundamentals of Digital Technology

Q1 **(a)** The overall level of response was very high with many candidates able to perform the required calculation. A significant number of candidates did not arrive at the correct final answer but still managed to achieve part marks by adhering to the requirement to show all their working.

(b) Although the majority of candidates could list the binary values as required, a significant number of candidates inexplicitly used four bits instead of two.

(c) This question discriminated as intended. Many excellent answers were provided. However, a common weakness was a failure to focus on the specific context of the question which referred to need for one and two bytes in ASCII and Unicode respectively.

(d) Both parts of this question discriminated rather more than intended. The question refers to sign and magnitude, but a common mistake was a failure to include the sign of the decimal value in Part (i).

Q2 **(a)** Most candidates were familiar with validation and verification in all three parts. In Part (iii), a significant number of candidates failed to make clear that the checking in double entry is an automatic process, not one carried out by the user.

(b) Most candidates were familiar with data and information and did refer to the shaded cell as required. Part (iii) discriminated as intended.

(c) While most candidates were familiar with the calculation of a modulus 11 check digit, the style of the question probed their knowledge effectively, at the same time reducing the amount of writing required.

Q3 **(a)** In general, the three terms were not properly understood with many candidates struggling to provide relevant technical detail. Port was better understood than either data bus or processor.

(b) This question discriminated as intended resulting in a wide range of marks. The style of the question probed the candidates' knowledge and understanding effectively, at the same time reducing the amount of writing required.

(c) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate a USB memory stick as a storage and transfer medium, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Q4 **(a)** In general, there was considerable confusion between the two types of operating system. Good quality answers were few and far between, with multitasking slightly better understood than multithreading which was poorly understood.

(b) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate CLI and WIMP from the point of view of the inexperienced user, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(c) Only a very small number of candidates were properly familiar with the two file formats to enable them to provide the required technical detail.

Q5 **(a)** Only a small number of candidates could distinguish between the WWW and the Internet in the concise, focused way that was expected.

(b) Most candidates were not properly familiar with the practical use of an Intranet by an organisation.

(c) Many candidates were familiar with HTML code to the required level and some complete responses to both question parts were provided. A number of candidates appeared to have been introduced to HTML code but they failed to describe the purpose of the lines in the given HTML code. A significant number of candidates resorted to guessing.

(d) Few candidates were familiar with the difference between client-side and server-side processing. Very few candidates could provide the required technical detail.

(e) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the decision of the business to move from proprietary software to open source software, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Assessment Unit A2 1 Information Systems

Q1 **(a)** The majority of candidates were familiar with the three types of network and could suggest an appropriate communication medium for each.

(b) Most candidates were familiar with the three network resources although wireless access point was not as well understood as the other two resources.

(c) The majority of candidates could complete the missing details and many complete responses were given.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate both topologies with respect to adding a new node, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Q2 **(a)** Most candidates could identify the four layers in the Transport Set of the OSI model.

(b) The majority of candidates were familiar to some extent with both communication protocols, only a small number of candidates could describe them in the appropriate detail.

(c) Only a very small number of candidates could distinguish between bandwidth and broadband as required.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate both mediums with respect to data transfer, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(e) The level of response to Part (i) and Part (ii) was very high. Part (iii) discriminated as intended. Only a small number of candidates were able to focus their answers to the key points in the scenario.

Q3 **(a)** This question discriminated as intended and the full range of marks was awarded. Some complete answers were provided but there was significant confusion between the requirements for 2NF and 3NF.

(b) The overall level of response was very high although many candidates struggled to frame their responses using the required DT terminology.

(c) The general standard of practical normalisation was very good and many complete responses were given. However, this report must stress again a continuing common failing. Candidates must be made aware that the entity names must be identified as well as the primary keys.

(d) The overall standard of the entity-relationship diagrams produced was very high.

(e) There were many excellent and succinct descriptions of the required changes.

Q4 **(a)** In Part (i) and Part (ii) there was clear evidence that most candidates were properly familiar with SQL. The overall level of response in Part (iii) was disappointing as many candidates chose to describe QBE rather than its use in producing the given query.

(b) Most candidates were properly familiar with neural networks and many complete responses were given.

(c) The majority of candidates could describe the role of the insurance consultants as required in Part (i).

The QWC question in Part (ii), purposely designed to identify the top candidates, required candidates to evaluate the expert system with respect to more accurate quotations, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Q5 **(a)** Most candidates could not identify the key differences between natural language processing systems and voice recognition systems.

(b) The overall level of response was good although cells were better described than base station controllers.

(c) Only a small number of candidates could relate the use of data mining to the privacy of supermarket customers.

(d) The overall level of response was very disappointing. Issues around cloud computing continue to cause problems to most candidates.

Q6 **(a)** In Part (i), the data subject was better understood than data controller, the latter term producing many confused and inaccurate descriptions.

In Part (ii), the majority of candidate demonstrated familiarity with the principles of Data Protection legislation but a significant number of candidates failed to describe these using appropriate language.

(b) The majority of candidate demonstrated familiarity the crimes identified in the Computer Misuse Act in Part (i), but a significant number of candidates failed to describe these using appropriate language.

Part (ii) concerned steps an organisation should take to ensure employees comply with the Computer Misuse Act. Most candidates failed to focus their response appropriately.

(c) This question referred to organisations addressing ethical issues in relation to digital technology. Most candidates failed to focus their response appropriately.

Principal Moderator's Report

Internal Assessment Overview

This is a piece of coursework submitted by Candidates based on a pre-released case study. In this series the case study was BelNote and it carries a 20% weighting of the overall A-level outcomes. Candidates spend 60 hours documenting and producing a relational database solution for BelNote. Centres assess the candidates work using assessment indicators and submit outcomes to CCEA. Moderation of centre outcomes then takes place.

Assessment Unit A2 2 Application Development

Unit Overview

The candidates carried out a problem solving exercise based on BelNote, the pre-released case study for the 2024 cohort. As in previous years the purpose of the case study is to give the candidates the opportunity to apply skills and knowledge, use Digital Technology to solve problems and document a solution. The case study encourages candidates to consider a full problem solving exercise , similar to solutions used in the previous series. It was good to see a variety of approaches to the BelNote Case Study. Some candidates approached it with the booking system route and others went down the purchasing and hiring/sales of musical instruments route. Most candidates took the booking of lessons approach as their solution. There is no requirement for the candidate to develop a Booking System and a Sales system.

The overall standard and quality of projects submitted as part of the moderation sample was very good and there was a good range of solutions linked to this particular case study. The moderation team are concerned about the increase in solutions that followed a teacher led template to problem solving within BelNote. It should be noted that candidates who follow a predefined template excludes an individual approach to system development and doesn't permit candidates to demonstrate their ability to solve problems.

Teachers should continue to be encouraged to follow the guidance given in the specification which clearly states that this task is a 60-hour assignment and carries a 20% weighting of the A level award. The application of the assessment criteria was very good in the majority of Centres and teachers demonstrated a clear understanding of using mark bands and actual mark points within the bands to make their judgements.

Teacher comments and annotation proves to be very useful during moderation as it helps moderators in coming to their decision. Very few centres in the 2024 series had no comments at all. Centres should also note the reporting back of marking by the moderation team on the TAC 6 report is also useful in preparation for assessing the next cohort. For the most part all centres marked accurately and according to the assessment criteria.

In the majority of centres, solutions were clearly labelled in electronic folders referenced by Candidate number identifying the top and bottom candidates on their electronic submission. Submission of Password protected databases is not required and can often cause issues during moderation, including access to the database solution. The majority of centres included the samples requested for Moderation in a well organised and well-presented manner.

The Analysis Section was done well by most centres. Candidates who were able to clearly identify and document their user requirements produced high quality overall solutions. User requirements more often than not were categorised into essential/non-essential. Candidates who only produced a limited number of requirements found it more difficult producing a detailed design of their solution. Gantt charts predominated project management tools with many candidates providing only mediocre levels of detail in the Gantt charts. It was disappointing to see data flow diagrams that lacked accuracy particularly with data flows. It should be stressed that Candidates should use the correct notation to produce their dataflow diagrams as outlined in the specification.

The design section seems to be understood well by candidates. Most candidates produced detailed Data Dictionaries which included appropriate data validation. The better candidates are fully explaining the different stages of Normalisation as they progress from 0NF to 3NF. It should be stressed that candidates should explain the stages their data is going through in the normalisation process. Candidates should also include ER diagrams that are linked to the normalised solution. The algorithm section has been tackled well with most candidates outlining the algorithms they plan to incorporate into their solution.

The inclusion of storyboards greatly assisted candidates in designing their solutions. Design of other components including forms, queries, macros and reports was generally done quite well. Candidates included evidence of designing a wide range of querying types for their solutions – append, delete, update and select. Report design was generally well done too with candidates designing sorting and grouping of data to show aggregate functions. Candidates who can produce successful solutions include a detailed test plan linked to user requirements. Test plans should closely be linked to user requirements.

In the Application Development stage the standard of relational database applications was generally quite high with many candidates utilising a wide range of querying and reporting features in their solutions. There were also deliberate attempts to enhance the user experience and provide a consistent thematic user interface across all menus and forms in the application. There was limited use of macros to implement processes and features across the samples.

As regards testing, most candidates are showing corrective action and fixes as a result of testing. Testing is approached well by most candidates, covering unit, system and integration. The majority of candidates were able to verify results of tests planned using the test plan created at an earlier stage.

User documentation seems to be fully understood by candidates as to what is expected. In the evaluation, most candidates were able to outline the strengths and limitations of their solutions together with ideas for extensions to the system, making reference to essential and non-essential user requirements. Some candidates created a login and password screen and there were no problems with this when the login and password was provided in a separate text file.

Contact details

The following information provides contact details for key staff members:

- **Specification Support Officer: Nuala Tierney**
  (telephone: (028) 9590 6689, email: ntierney@ccea.org.uk)
- **Officer with Subject Responsibility: Andrew Douglas**
  (telephone: (028) 9590 6701, email: adouglas@ccea.org.uk)

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COUNCIL FOR THE CURRICULUM, EXAMINATIONS AND ASSESSMENT
29 Clarendon Road, Clarendon Dock, Belfast BT1 3BG
Tel: +44(0)28 9026 1200 Email: info@ccea.org.uk
Web: ccea.org.uk

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